

Analyzing the Impact of E-sports Tournaments on the Steam Community Market: A Case Study on CS2 Skins and Stickers

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Project Website: <https://serkandag42.github.io/Dsa210-Term-Project/>

1. Abstract

The rapid expansion of the esports industry has transformed in-game virtual cosmetics into highly liquid, structured financial commodities. This research investigates the macroeconomic relationship between Counter-Strike 2 (CS2) professional esports viewership on Twitch and the internal digital asset market on the Steam platform. Utilizing a historical dataset encompassing market transactions, S-Tier tournament schedules, and daily streaming metrics, this study applies both inferential statistics and machine learning frameworks. The findings confirm that while exact daily asset pricing is bounded by the Random Walk Hypothesis, esports crowd density remains the absolute dominant macroeconomic catalyst driving market volatility and structural economic states.

2. Overview & Motivation

Modern gaming ecosystems feature internal economies with volatility profiles comparable to traditional asset derivatives boards. Within CS2, the community market dictates the value of virtual assets purely based on supply, demand, and public hype.

As outlined in the initial proposal, the objective of this project is to quantify how high-stakes competitive events, such as Majors and ESL tournaments, function as catalysts for price volatility in in-game items. By bridging the gap between social engagement and digital finance, this project aims to understand these "Hype Factors," which are crucial for modeling digital commodity markets driven by community sentiment and scarcity.

3. Data Source & Collection Methodology

To ensure robust empirical evaluation, independent data streams were integrated using custom Python pipelines:

- **Steam Community Market Data:** Historical transaction prices and sales volumes of representative Tier-1 CS2 weapon skins and tournament stickers were retrieved using the Steam Web API and custom Python-based scraping tools. The individual item prices were aggregated into a singular, normalized "Market Index" to capture macroeconomic shifts.
- **Tournament Metrics:** Historical match schedules, event timelines, and tournament tiers (e.g., PGL Copenhagen, Paris Major) were sourced from HLTV.org and Liquipedia using BeautifulSoup.
- **Viewership Data:** A Python-based tracking script was utilized to capture historical concurrent viewership numbers from Twitch/YouTube to measure the impact of audience size on immediate market demand.

All timelines were synchronized, handling missing values through strict chronological alignment. Data variables underwent log-transformations to resolve severe distribution skewness and were normalized using Scikit-Learn's StandardScaler.

4. Exploratory Data Analysis (EDA) & Hypothesis Testing

Before constructing predictive models, foundational assumptions were tested through rigorous statistical frameworks and visualizations:

- **Pearson Correlation:** The analysis identified a statistically significant, non-linear positive relationship between active viewing density and market value indices ($p < 0.05$).
- **Two-Sample T-Test:** The evaluation confirmed that transaction indices during operational championship dates significantly deviate from non-participating or non-tournament baselines, proving that competitive windows inject volatility.
- **One-Way ANOVA:** Variance testing verified that distinct competitive layers execute varying scales of economic impacts on the asset ecosystem.

5. Machine Learning Methodology

The project deployed a dual-phase machine learning architecture, combining unsupervised market segmentation with supervised regression benchmarking to predict price movements.

5.1. Unsupervised Learning: Market State Clustering

To extract structural behaviors from fluctuating economic periods, K-Means clustering was applied. The model autonomously partitioned the CS2 marketplace grid into three distinct macroeconomic states:

1. **Stagnant Market:** Characterized by lower baseline community interaction and standard asset indices.
2. **Transition Phase:** Intermediate nodes showcasing pricing stickiness and gradual hype buildup.
3. **Hype Market:** Explosive tournament streaming volumes driving aggressive market expansion.

5.2. Supervised Learning: Predictive Regression

Four distinct regression frameworks were benchmarked: Linear Regression, K-Nearest Neighbors (mathematically optimized via GridSearchCV), Random Forest, and Gradient Boosting Regressors. Autoregressive variables (e.g., lagged historical prices) and artificial time trends were strictly excluded from the feature space to prevent chronological "Target Leakage," ensuring the models evaluated genuine external impacts.

6. Empirical Results & Discussion

The supervised predictive models yielded baseline R^2 parameters approximating zero (or slightly negative, e.g., Random Forest) across all independent variables. In the context of quantitative finance, this matches the **Random Walk Hypothesis**. Attempting to predict the exact, continuous absolute value of an entire marketplace utilizing single external popularity metrics is fundamentally bounded by market efficiency.

However, despite the low linear predictability of the overarching index, the **Feature Importance Profile** extracted from the tree-based algorithms provided definitive empirical proof of the primary thesis. Daily Twitch audience volume (avg_viewer) accounted for over 80% of the structural node-split decision weight. This verifies that daily streaming crowd density is the absolute dominant vector driving value variance, significantly outperforming static tournament calendar slots.

7. Conclusions & Future Work

This data science pipeline confirms that the CS2 virtual marketplace functions as a complex financial system. While predicting continuous asset indices directly from social metrics presents high non-linear difficulties, esports crowd density remains the essential driver of market behavior.

Future Work: Subsequent iterations of this research could heavily benefit from integrating Natural Language Processing (NLP) models to compute real-time Sentiment Analysis metrics from platforms like Reddit and X (Twitter), capturing the immediate emotional context of the community alongside raw viewership volumes.